CAPSTONE PROJECT REPORT

InsightCommerce

An AI-Powered E-Commerce Analytics and Decision Intelligence Platform

Prepared by

Aisha Afroj

Submission date

17 August 2026

Dataset	Synthetic E-Commerce Electronic Sales 2025
Source	Kaggle - Wojciech Kie■bowicz
Verified size	108,300 transactions x 10 source fields
Application	Streamlit + DuckDB + Plotly + scikit-learn
AI query modes	Ollama, optional OpenAI, offline demo fallback

Submission evidence is reproducible from the repository: pinned dependencies, 21 passing tests, a 10/10 query benchmark, regular Git commits, and measured filtered aggregation latency below 3 ms on the development Mac.

Abstract and project result

InsightCommerce is an end-to-end analytics application for a verified Kaggle dataset of 108,300 synthetic global electronics transactions from 2025. The system validates the source schema, creates a quality profile, derives transparent calendar and product-taxonomy features, stores optimized Parquet, and serves filtered aggregation through DuckDB. A schema-aware assistant translates natural-language questions to a strict JSON query plan and read-only SQL. An AST safety gate blocks mutation, external access, sensitive-shaped fields, unrestricted star selection, excessive rows, and long execution; one corrective retry and five-turn memory support reliable conversation. The Streamlit interface provides global filters, preset insights, eight visualization types, AI chart recommendation with manual override, and PDF, DOCX, CSV, HTML, and PNG export paths. Advanced features include calendar-based daily demand prediction and Isolation Forest anomaly detection. Evaluation recorded 21 passing automated tests, 10/10 benchmark questions, a 2.52 ms median filtered aggregation, and a demand-model held-out R-squared of 0.995. The result is reproducible, deployment-ready, and deliberately transparent about the synthetic data and model limits.

Measured completion summary

Evidence	Result	Acceptance
Source integrity	108,300 rows; 0 missing; 0 duplicate IDs	PASS
Filtered aggregation	Median 2.52 ms	PASS (<500 ms)
Automated verification	21 tests + static quality checks	PASS
Question benchmark	10 of 10 questions	100%
Dashboard	5 sections; 8 prepared chart types	PASS
Advanced features	Demand prediction + anomaly detection	PASS

Central design choice: adapt to the real ten-column file. Profit, discount, shipping, and cancellation are not present, so no synthetic analytical field was invented.

1. Introduction, objectives, and dataset scope

Modern analytics users expect to move from a business question to a trustworthy result without manually writing code. The project problem is therefore broader than a dashboard: it requires a reproducible data layer, a bounded AI code-generation path, interactive visual reasoning, exportable evidence, and advanced analytics that remain honest about the dataset's limits.

Objectives

- Verify the downloaded file before implementation and preserve the original unchanged.

- Deliver fast filtered aggregation, schema inspection, and measurable data-quality evidence.

- Convert natural language to safe DuckDB SQL with bounded retry and conversational context.

- Provide at least six interactive visual forms plus AI selection and human override.

- Add two advanced features: demand prediction and anomaly detection.

- Package tests, deployment configuration, documentation, benchmarks, final report, and ZIP.

Dataset scope

Dimension	Verified value
Period	2025-01-01 to 2025-12-31
Markets	15 countries
Products	105 electronics products
Customers	10,830 synthetic customer identifiers
Order value	$9.99 to $16,191.00; exact price x quantity
Total simulated revenue	$133,323,271.88

Although names and email addresses are synthetic, they are treated as sensitive-shaped data. They remain in the preserved source but are excluded from prompts, general exports, anomaly results, and the generated-query allowlist.

2. System architecture

The architecture separates trusted preparation, interactive presentation, generated-code control, and analytical outputs. DuckDB receives either parameterized application filters or AST-validated generated SQL. No LLM receives raw customer names or email addresses.

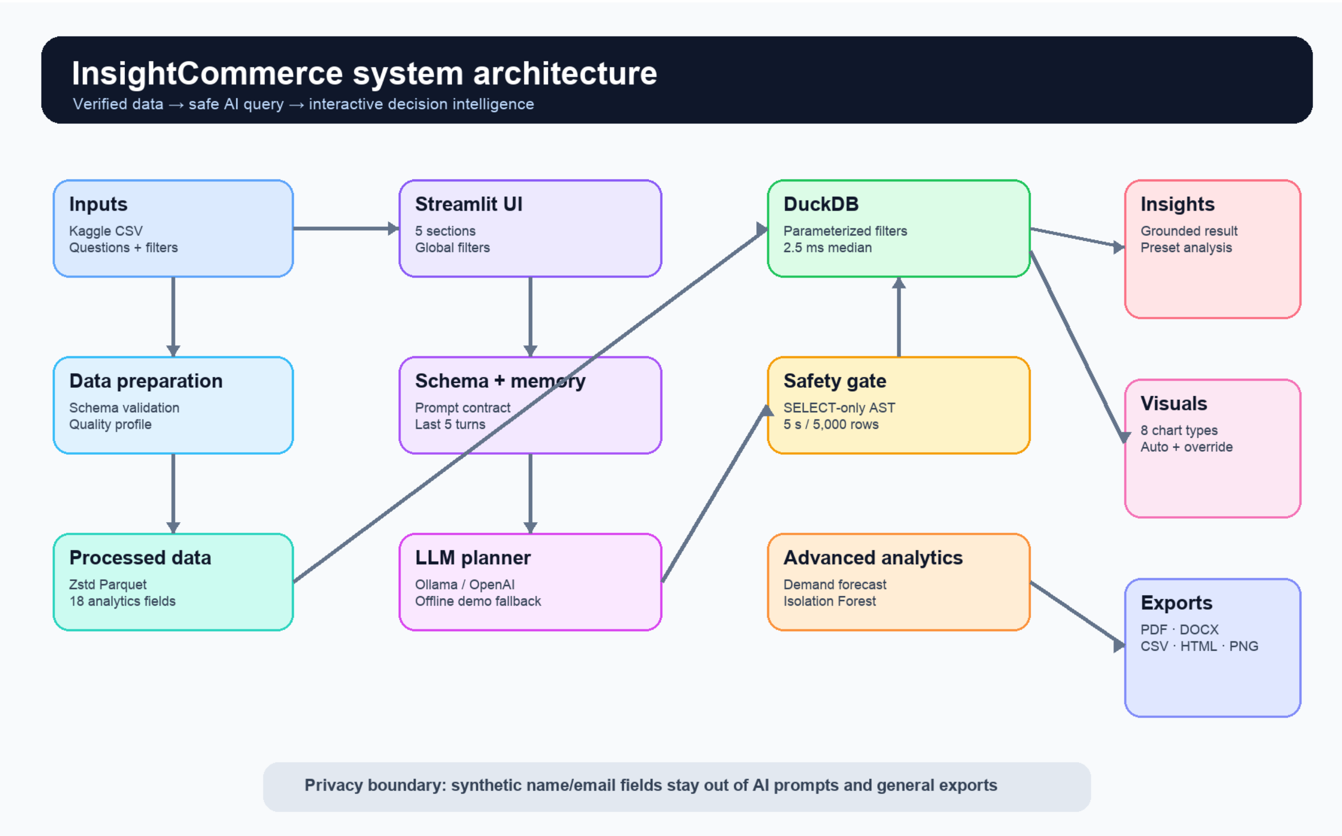

Figure 1. InsightCommerce component and trust-boundary architecture.

Layer	Responsibility	Primary evidence
Data	Validate, enrich, profile, convert, aggregate	quality_report.json; performance.json
AI query	Schema prompt, JSON plan, retry, memory	prompts.py; assistant.py
Safety	Parse, allowlist, cap, timeout, disable external access	sandbox.py; security tests
Experience	Filters, charts, explanations, exports	app.py; browser QA
Advanced	Demand forecast and anomaly ranking	ml.py; anomaly.py; model card

3. Task A - data backend and performance

The raw ZIP checksum is 81a80e884aeaf28f6816af25fa334dbcd712716337208194c5b451be339b63db. The preparation command verifies exact column order, parses dates and numerics, strips identifier whitespace, enforces unique order IDs, confirms price x quantity, derives calendar/taxonomy fields, and writes Zstd-compressed Parquet plus JSON metadata.

Field	Type	Analytical use
order_id	string	Unique order count
date	date	Trend and seasonality
customer_id	string	Repeat-customer count
customer_name	string	Excluded from AI/general exports
customer_email	string	Excluded from AI/general exports
country	string	Geographic comparison
product	string	Product ranking and taxonomy
price	float	Unit-price distribution
quantity	integer	Units and transaction pattern
order_value	float	Simulated revenue

Performance result: 30 warm filtered aggregations produced a 2.52 ms median, 2.66 ms p95, and 2.71 ms maximum - comfortably below the 500 ms target.

4. Task B - schema-aware AI and safe text-to-code

Each question is combined with the public schema catalog and at most five previous successful turns. The model must return a strict QueryPlan containing SQL, chart type, result field mapping, title, and rationale. The SQL is never executed directly.

Step	Control	Failure behavior
1	Question + schema + recent memory	No raw PII sample is sent
2	Strict JSON Schema query plan	Pydantic rejects missing/extra fields
3	sqlglot AST allowlist	Unsafe plans are blocked before DuckDB
4	5 s timeout + 5,000-row wrapper	Query is interrupted/capped
5	One error-informed corrective retry	Second failure becomes a clear user error
6	Grounded result narration	Only executed result values are described

Provider strategy

Mode	Purpose	Credential
Ollama	Local live LLM demonstration	None; local model required
OpenAI Responses	Optional hosted Structured Outputs	OPENAI_API_KEY
Offline demo	Deterministic common-question evaluation	None; clearly labeled fallback

Security tests prove that DROP, DELETE, COPY, external CSV reads, hidden schema access, multiple statements, sensitive fields, and unrestricted star selection are rejected.

5. Task C - interactive application

The Streamlit application is organized into Overview, Visual Explorer, AI Analyst, Prediction & Anomalies, and Reports & Quality. Date, country, and product-category filters apply consistently across the dashboard. The default overview remains legible at a tested 1600 x 1000 desktop viewport with no horizontal overflow or application exceptions.

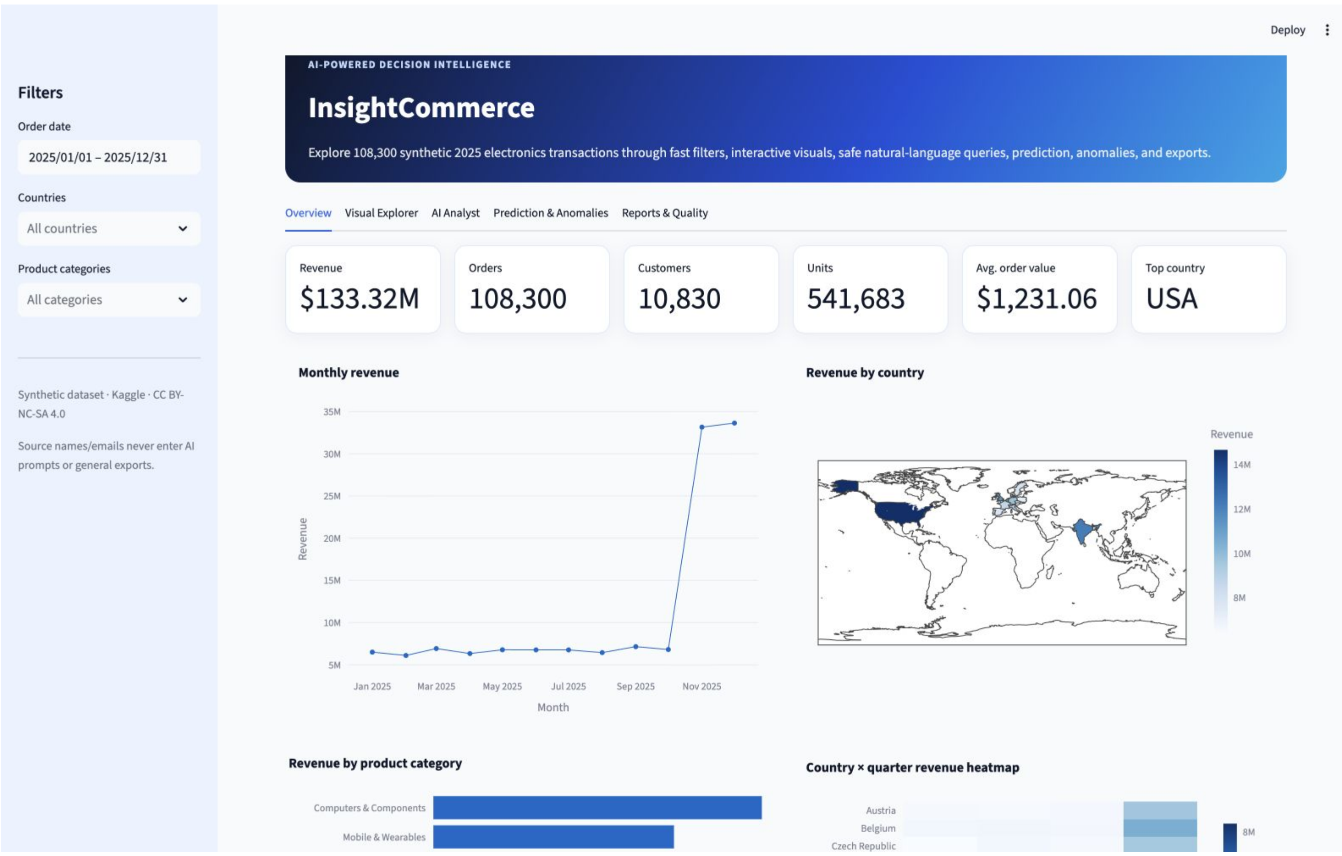

Figure 2. Browser-verified default dashboard on the development Mac.

Section	User outcome
Overview	KPIs, trend, map, category comparison, quarter heatmap
Visual Explorer	Eight prepared visual forms plus manual builder and chart downloads
AI Analyst	Question -> safe SQL -> table -> chart -> grounded explanation
Prediction & Anomalies	Forecast a date and investigate unusual orders
Reports & Quality	Inspect schema/quality and download PDF/DOCX summaries

6. Visualization and analytical evidence

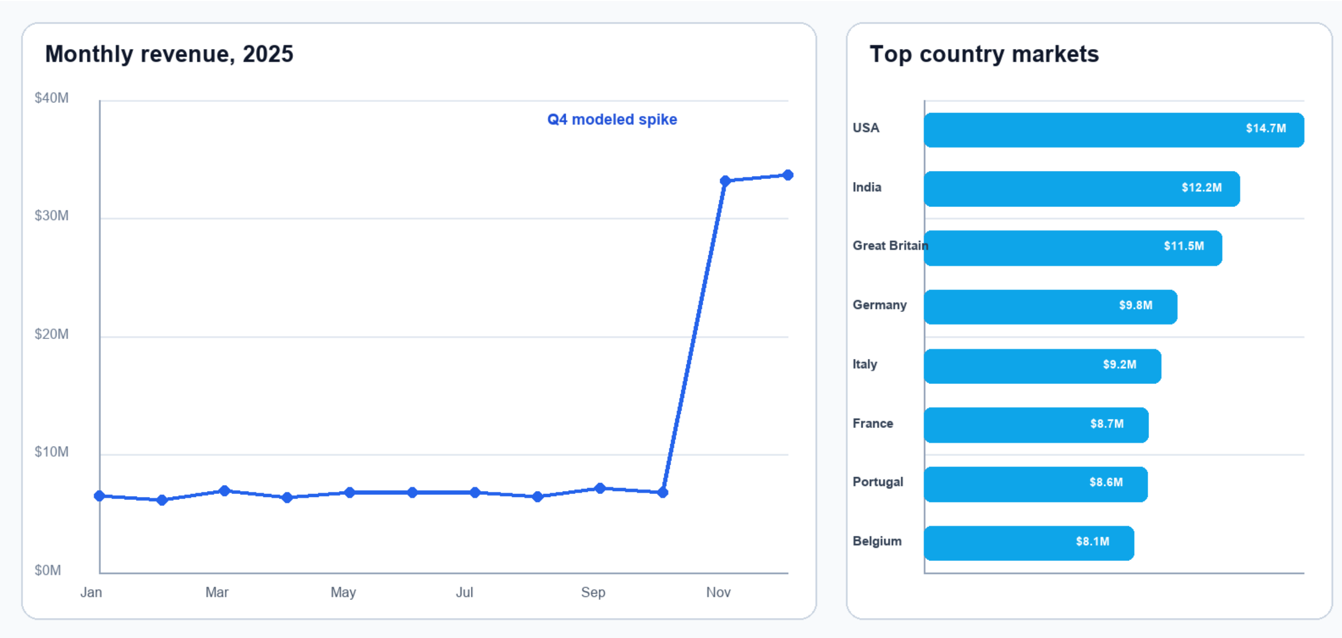

Figure 3. Measured 2025 revenue pattern and strongest country markets.

The source intentionally weights Q4. Revenue is roughly $6-7 million per month from January through October, then rises above $33 million in both November and December. This is a modeled pattern, not evidence about a real company. USA leads total revenue because the synthetic generator also assigns different country record volumes.

Visual	Purpose	Interactivity
Line	Monthly revenue trend	Hover, zoom, range inspection
Choropleth	Country revenue	Country hover and scale
Bar	Category ranking	Sorted comparison
Heatmap	Country x quarter concentration	Color/intensity inspection
Treemap	Category -> product hierarchy	Drill and hover
Scatter	Price, quantity, order value	Multi-encoding exploration
Box	Price distribution by category	Outlier visibility
Histogram	Order-value distribution	Binned frequency

AI chart hints are accepted only when referenced columns and types are compatible. The user can override chart type and fields; incompatible combinations fall back safely.

7. Task D feature 1 - predictive analytics

Figure 4. Daily order-demand fit and model feature importance.

Metric	Held-out result	Interpretation
Test design	Every seventh day (53 days)	Seasonally distributed holdout
MAE	12.4 orders/day	Average absolute error
RMSE	18.2 orders/day	Penalizes larger misses
R-squared	0.995	Strong fit to synthetic seasonality
Model	Random Forest; seed 42	Reproducible nonlinear calendar model

Feature and prediction design

Known-in-advance features are month, quarter, weekday, day of month, annual sine/cosine, weekend, and Q4. The interface predicts daily order count for a chosen date. Indicative revenue is then predicted orders multiplied by the historical median order value; it is explicitly labeled as an assumption rather than a separate learned target.

The high R-squared reflects the dataset's designed Q4 step change. It must not be generalized to real retail demand without multi-year real data and rolling-origin tests.

8. Task D feature 2 - anomaly detection

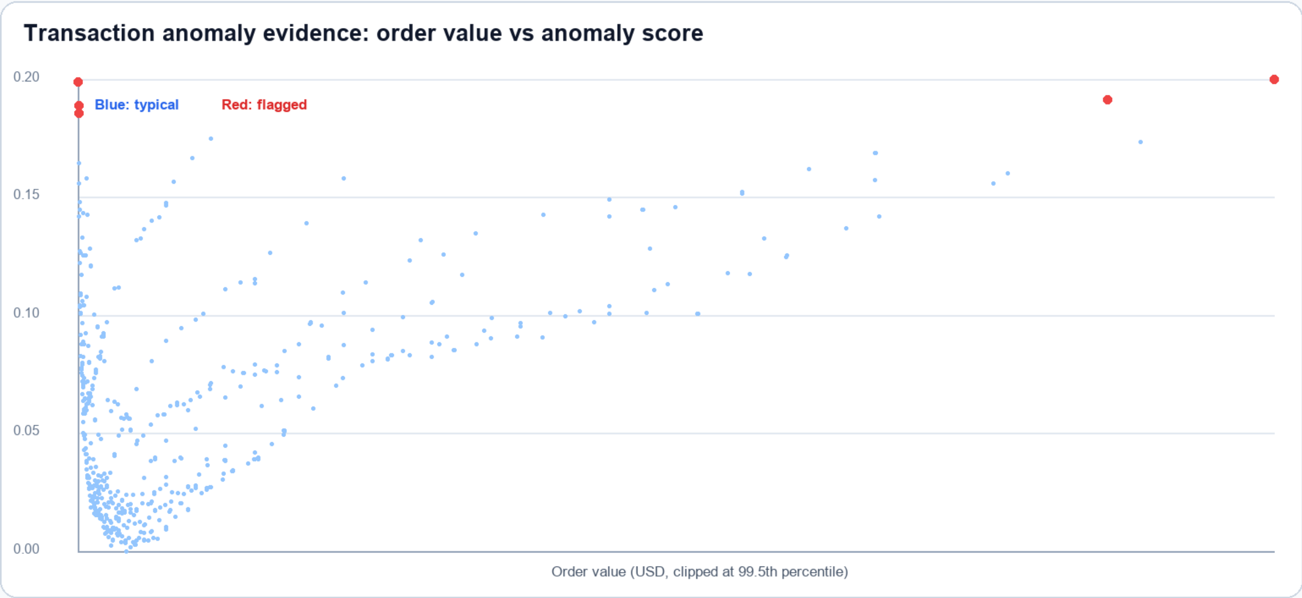

Figure 5. Isolation Forest anomaly scores with flagged points in red.

Element	Implementation
Model	Isolation Forest, 160 trees, seed 42
Features	log price, quantity, log order value, value / product median
Nominal contamination	1%
Observed flags	1,006 of 108,300
Observed rate	0.93%
Output	score, flag, and transparent reason code

Interpretation boundary

Reason codes identify exceptionally high order value, premium unit price, maximum quantity, values well above a product's median, or a broader unusual multivariate combination. The output excludes name and email. A flag is an investigative signal - not proof of fraud, data error, or misconduct. Score ties can make the observed rate slightly different from the configured contamination.

9. Evaluation, testing, and browser QA

Verification area	Evidence	Result
Data contract	Exact schema, shape, identity checks	PASS
Generated SQL safety	7 malicious/unsafe cases + safe COUNT(*)	PASS
Retry behavior	Unsafe first plan, valid second; no third	PASS
Conversation memory	Seven writes retain only latest five	PASS
Charts	Time/country inference and invalid-hint fallback	PASS
Prediction/anomaly	Finite metrics, output, privacy columns	PASS
Exports	PDF, DOCX, CSV, interactive HTML signatures	PASS
Performance	Median 2.52 ms vs 500 ms	PASS
Static quality	Ruff across app, source, scripts, tests	PASS
Total automated tests	21	PASS

Browser walkthrough

The application was launched on the development Mac and exercised through the browser. All five primary tabs and both nested advanced/report tabs were opened. The offline monthly AI question executed in one attempt at 6.70 ms. An initial duplicate Plotly element ID was observed between hidden tabs, fixed with unique keys, and reverified. Final snapshots showed zero Streamlit exception elements and no horizontal overflow at 1600 x 1000.

Evaluation principle: pass/fail claims in this report come from committed JSON/Markdown artifacts, automated tests, or the final browser walkthrough.

10. Ten-question accuracy benchmark

The benchmark exercises the full offline question path: natural-language plan selection, strict plan parsing, SQL safety validation, DuckDB execution, chart compatibility, and grounded narration. Each result is compared with independently executed reference SQL.

#	Question	Attempts	Time	Status
1	What is the total revenue?	1	0.65 ms	PASS
2	How many orders are there?	1	0.35 ms	PASS
3	How many units were sold?	1	0.48 ms	PASS
4	Show the monthly revenue trend for 2025.	1	2.42 ms	PASS
5	Which countries generated the most revenue?	1	2.49 ms	PASS
6	Which country has the highest average order value?	1	2.23 ms	PASS
7	Show the top 10 products by revenue.	1	2.90 ms	PASS
8	Compare revenue across product categories.	1	2.40 ms	PASS
9	Compare every quarter and highlight Q4 performance.	1	1.08 ms	PASS
10	What are the leading products by revenue?	1	2.73 ms	PASS

Per-question criterion	Required
Exact numeric/tabular result	Yes
Required result columns	Yes
Compatible chart recommendation	Yes
Success without corrective retry	Yes
Non-empty grounded narrative	Yes

Final benchmark accuracy: 10/10 (100%). Live LLM accuracy remains provider/model dependent and should be evaluated separately before production use.

11. Security, privacy, and reliability

Risk	Control	Residual limitation
Prompt injection	Question/history labeled untrusted; system contract fixed	A model may still propose bad SQL; validator is authoritative
SQL mutation	AST permits one SELECT/CTE only	Parser/library updates require regression tests
File/network access	Functions blocked; DuckDB external access disabled	OS/container permissions remain defense-in-depth
Sensitive-shaped fields	Prompt schema omits name/email; validator rejects them	Raw synthetic source still contains these columns
Resource exhaustion	5 s interrupt, 5,000 rows, 1 GB DuckDB limit	Host-level limits should also be configured
Hallucinated insight	Narrative is generated from executed result values	Live-model explanatory text still needs user judgment
Secret exposure	Environment/Streamlit secrets; real files ignored by Git	Operators must rotate compromised keys

Reliability controls

- Fixed random seeds make prediction and anomaly results reproducible.
- Parameterized application filters keep user values out of SQL strings.
- Only successful answers enter conversation memory.
- AI chart hints are type-checked before rendering.
- The original CSV and ZIP remain unchanged and attributable under their own license.
- CI reruns data preparation, tests, static checks, and both benchmarks.

12. Reproducibility and deployment preparation

A clean environment can reproduce the application with the pinned requirements and included raw data. Generated Parquet and quality files are intentionally rebuilt.

```
python3.12 -m venv .venv
source .venv/bin/activate
pip install -r requirements-dev.txt
python scripts/prepare_data.py
pytest
python scripts/benchmark_performance.py
python scripts/run_question_benchmark.py
streamlit run app.py
```

Prepared deployment targets

Target	Included support	AI behavior
Local macOS	Virtual environment + Streamlit	Ollama, OpenAI, or offline demo
Streamlit Cloud	app.py, runtime.txt, pinned requirements	Offline demo or hosted secret
Docker	Dockerfile + health check	Configure reachable protected provider
Process host	Procfile	Provider via environment variables
GitHub CI	Test, lint, data prep, two benchmarks	No external LLM required

Version control

Development is divided into regular milestone commits: verified data backend; safe schema-aware AI engine; dashboard and advanced analytics; submission documentation and evaluation; and final report/package. This avoids a prohibited single-commit submission and makes review traceable.

13. Limitations, future work, conclusion, and references

Limitations

- The source is synthetic and represents only 2025; findings are demonstrations, not company facts.
- Order value is deterministically price x quantity; no profit, cost, discount, shipping, or returns exist.
- Country volumes and Q4 behavior were modeled by the dataset creator and drive several results.
- The forecast holdout is seasonally distributed, not a multi-year rolling-origin production test.
- Isolation Forest identifies unusual combinations, not fraud or operational error.
- Live LLM results depend on provider availability, model behavior, latency, and cost.

Future work

With instructor approval and a richer real dataset, future versions should add product cost/profit, discount, shipping, return status, and customer segment fields; evaluate rolling multi-year forecasts; add authenticated user roles and audit logs; benchmark multiple live LLMs; and deploy managed observability with rate limits and query traces.

Conclusion

InsightCommerce fulfills Tasks A-D as an integrated, testable application. Its strongest contribution is not a single chart or model, but the controlled chain from verified data through generated code to a grounded, exportable result. The final system is fast, visually complete, privacy-conscious, reproducible, and explicit about what the dataset can and cannot support.

References

1. W. Kielbowicz, Synthetic E-Commerce Electronic Sales 2025 Dataset, Kaggle, 2026. https://www.kaggle.com/datasets/wojciechkiebowicz/e-commerce-electronic-sales-2025-dataset

2. DuckDB Documentation. https://duckdb.org/docs/

3. Streamlit Documentation. https://docs.streamlit.io/

4. Plotly Python Documentation. https://plotly.com/python/

5. scikit-learn User Guide: Random Forests and Isolation Forest. https://scikit-learn.org/stable/user_guide.html

6. OpenAI API Documentation: Structured model outputs. https://developers.openai.com/api/docs/guides/structured-outputs

7. Ollama Documentation. https://docs.ollama.com/

8. sqlglot documentation and source. https://github.com/tobymao/sqlglot